

Linear Equations & Systems



Solving single-variable linear equations

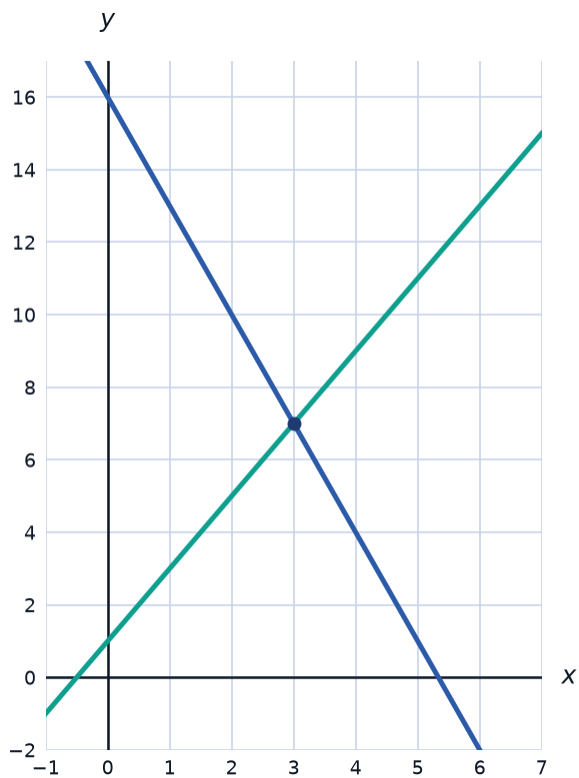
- 1 Solve for x : $3x - 7 = 20$.
- 2 Solve for x : $\frac{2x + 5}{3} = 7$.
- 3 Solve for x : $5(x - 2) = 3(x + 4)$.

Equations with no solution or infinitely many solutions

- 4 For what value of k does the equation $4x + 6 = kx + 6$ have infinitely many solutions?
- 5 Explain why the equation $2x + 3 = 2x + 7$ has no solution, without solving it algebraically first.

Solving systems by substitution

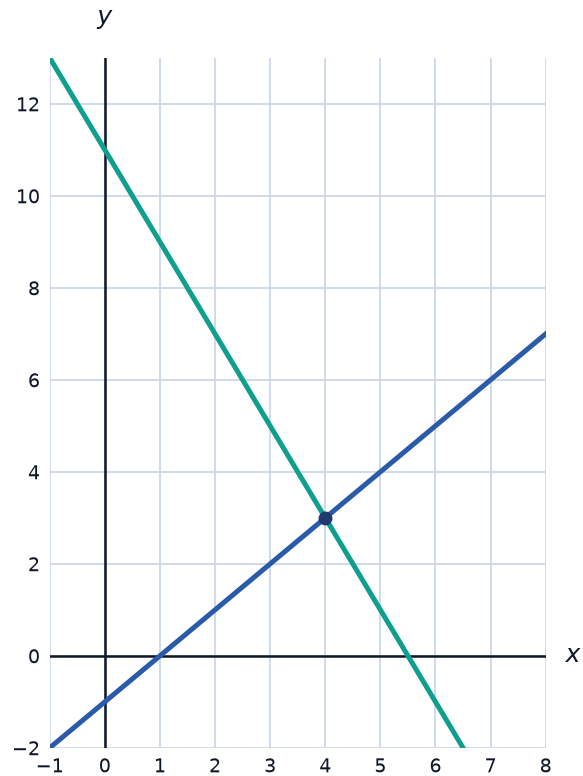
- 6 Solve the system: $y = 2x + 1$ and $3x + y = 16$.



- 7 Solve the system: $x - 2y = 5$ and $2x + 3y = 3$.

Solving systems by elimination

- 8 Solve the system: $2x + y = 11$ and $x - y = 1$, using elimination.

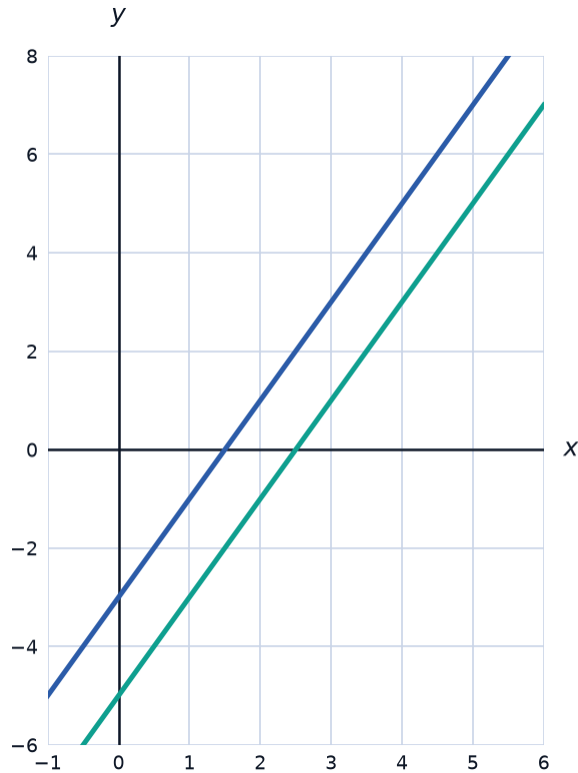


- 9 Solve the system: $3x + 2y = 16$ and $5x - 2y = 8$, using elimination.

Systems with no solution or infinitely many solutions

- 10 For what value of c does the system $6x + 9y = 12$ and $2x + 3y = c$ have infinitely many solutions?

- 11 Explain why the system $4x - 2y = 10$ and $2x - y = 3$ has no solution.



Translating word problems into linear equations

- 12 A rectangle's length is 4 more than twice its width. If the perimeter is 68, find the width and length.
- 13 A taxi charges a flat fee of 4 plus 2.50 per mile. If a ride costs 21.50, how many miles was the ride?

Systems as word problems

- 14 A movie theater sells adult tickets for 12 and child tickets for 8. If 150 tickets were sold for a total of 1560, how many of each type were sold?
- 15 Two numbers have a sum of 45 and a difference of 11. Find the two numbers.

Interpreting the slope and intercept of a linear model

- 16 A gym membership costs $C = 25 + 15m$ dollars, where m is the number of months. Interpret the meaning of the numbers 25 and 15 in this context.

- 17 Two companies charge for printing: Company A charges $C = 0.10p + 20$ and Company B charges $C = 0.15p$, where p is the number of pages. Find the number of pages at which both companies charge the same amount, and state which company is cheaper for 500 pages.
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Equations with variables on both sides and fractions

- 18 Solve for x : $\frac{x}{3} + \frac{x}{4} = 7$.
- 19 Solve for x : $2(x + 3) - 5 = 3(x - 1) + 2$.
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Systems with three unknowns represented as a word problem

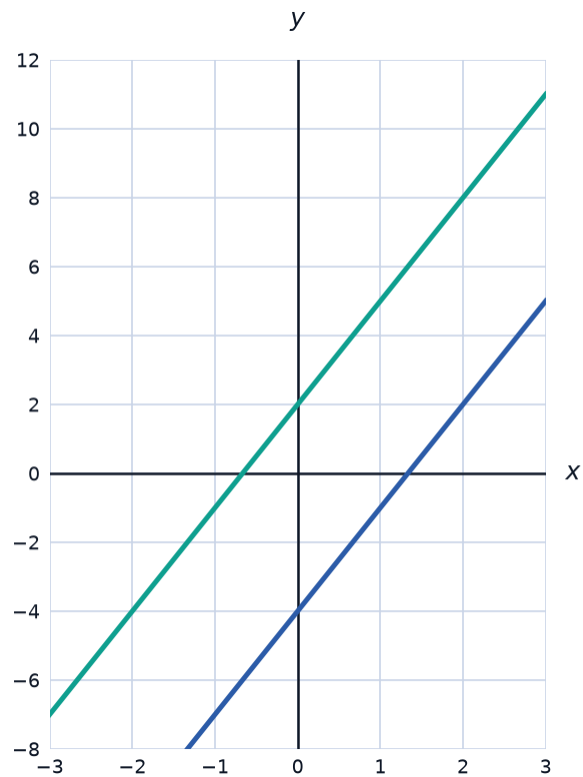
- 20 A vending machine sells snacks for 1.50 dollars and drinks for 2 dollars. On one day it sold 45 items total for 79.50 dollars. Set up and solve a system to find the number of snacks and drinks sold.
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Choosing the best method for a given system

- 21 For the system $y = 4x - 3$ and $5x + 2y = 17$, explain which method (substitution or elimination) is more efficient here, and solve the system.
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Consistent, inconsistent, and dependent systems

- 22 Classify the system $y = 3x + 2$ and $y = 3x - 4$ as consistent, inconsistent, or dependent, and explain your reasoning without solving.



Multiple choice — easy 15

- 23 Solve for x : $4x + 2 = 8x - 58$.

- a. -15
- b. 16
- c. 15
- d. 14

24 Solve for x : $-7x + 13 = -8x + 7$.

a. -6

b. -5

c. 6

d. -7

25 Solve for x : $-6x + 10 = 9x - 5$.

a. 1

b. 2

c. -1

d. 0

26 Solve for x : $8x - 10 = 3x - 80$.

a. 14

b. -13

c. -14

d. -15

27 Solve for x : $2x + 15 = -8x - 95$.

a. -10

b. -11

c. -12

d. 11

28 Solve for x : $9x - 19 = 7x - 37$.

- a. 9
- b. -10
- c. -8
- d. -9

29 Solve for x : $9x - 19 = -6x - 4$.

- a. 1
- b. 2
- c. -1
- d. 0

30 Solve for x : $x - 10 = -8x + 116$.

- a. 14
- b. -14
- c. 15
- d. 13

31 Solve for x : $-8x + 14 = x - 13$.

- a. 3
- b. -3
- c. 2
- d. 4

32 Solve for x : $3x - 19 = 8x - 64$.

- a. 8
- b. 9
- c. -9
- d. 10

33 Solve for x : $8x - 4 = -2x - 154$.

- a. -15
- b. -14
- c. -16
- d. 15

34 Solve for x : $-8x - 3 = -5x - 24$.

- a. 8
- b. -7
- c. 6
- d. 7

35 Solve for x : $-x + 16 = 3x - 40$.

- a. 14
- b. 13
- c. 15
- d. -14

36 Solve for x : $-3x + 9 = 3x + 75$.

- a. 11
- b. -12
- c. -11
- d. -10

37 Solve for x : $7x + 1 = -3x + 51$.

- a. 5
- b. 4
- c. 6
- d. -5

Multiple choice — medium 15

38 Solve the system: $-2x + 6y = 22$ and $x + 6y = -2$.

- a. $(-8, 1)$
- b. $(-8, 2)$
- c. $(-7, 1)$
- d. $(1, -8)$

39 Solve the system: $0x + 3y = -3$ and $5x + 5y = -45$.

- a. $(-8, -1)$
- b. $(-1, -8)$
- c. $(-8, 0)$
- d. $(-7, -1)$

40 Solve the system: $2x - 3y = 11$ and $0x - 6y = -18$.

a. (10, 4)

b. (3, 10)

c. (10, 3)

d. (11, 3)

41 Solve the system: $-2x + 5y = -24$ and $-3x - 2y = -17$.

a. (-2, 7)

b. (7, -2)

c. (8, -2)

d. (7, -1)

42 Solve the system: $-3x - y = 35$ and $-5x - 3y = 65$.

a. (-9, -5)

b. (-10, -4)

c. (-5, -10)

d. (-10, -5)

43 Solve the system: $-6x - 5y = -4$ and $3x - 2y = 47$.

a. (9, -9)

b. (9, -10)

c. (-10, 9)

d. (10, -10)

44 Solve the system: $4x + 5y = 51$ and $-4x - 2y = -30$.

a. (5, 7)

b. (4, 8)

c. (7, 4)

d. (4, 7)

45 Solve the system: $-x + 6y = -44$ and $4x + 6y = -34$.

a. (2, -7)

b. (3, -7)

c. (-7, 2)

d. (2, -6)

46 Solve the system: $-4x - 2y = -4$ and $-2x - 5y = 38$.

a. (6, -10)

b. (7, -10)

c. (6, -9)

d. (-10, 6)

47 Solve the system: $x + 3y = 11$ and $-x + 6y = 34$.

a. (5, -4)

b. (-4, 6)

c. (-3, 5)

d. (-4, 5)

48 Solve the system: $-2x - 6y = 28$ and $2x - 4y = 42$.

- a. $(-7, 7)$
- b. $(7, -7)$
- c. $(8, -7)$
- d. $(7, -6)$

49 Solve the system: $0x - y = 3$ and $5x - 4y = 57$.

- a. $(9, -3)$
- b. $(-3, 9)$
- c. $(9, -2)$
- d. $(10, -3)$

50 Solve the system: $-4x - 4y = -40$ and $-4x + 4y = 40$.

- a. $(10, 0)$
- b. $(1, 10)$
- c. $(0, 11)$
- d. $(0, 10)$

51 Solve the system: $3x + 3y = -21$ and $-5x - 2y = 35$.

- a. $(-7, 0)$
- b. $(-7, 1)$
- c. $(-6, 0)$
- d. $(0, -7)$

52 Solve the system: $x + 2y = -12$ and $5x + 6y = -44$.

a. $(-4, -3)$

b. $(-4, -4)$

c. $(-5, -2)$

d. $(-3, -4)$

Multiple choice — hard 15

53 For what value of c does the system $10x + 20y = 25$ and $2x + 4y = c$ have infinitely many solutions?

a. 10

b. 5

c. 0

d. 25

54 For what value of c does the system $-15x - 5y = -25$ and $-3x - y = c$ have infinitely many solutions?

a. -5

b. -10

c. -25

d. 0

55 For what value of c does the system $-15x - 30y = -60$ and $-3x - 6y = c$ have infinitely many solutions?

a. -17

b. -7

c. -12

d. -60

- 56 For what value of c does the system $0x - 6y = -28$ and $0x - 3y = c$ have infinitely many solutions?
- a. -28
 - b. -12
 - c. -16
 - d. -14
- 57 For what value of c does the system $-8x - 12y = 44$ and $-2x - 3y = c$ have infinitely many solutions?
- a. 7
 - b. 15
 - c. 44
 - d. 11
- 58 A store sells swim passes for 20 dollars and gym passes for 12 dollars. If 82 items were sold for a total of 1072 dollars, how many of each type were sold?
- a. $(71, 11)$
 - b. $(6, 76)$
 - c. $(11, 71)$
 - d. $(16, 66)$
- 59 A store sells burgers for 17 dollars and salads for 13 dollars. If 136 items were sold for a total of 2012 dollars, how many of each type were sold?
- a. $(56, 80)$
 - b. $(61, 75)$
 - c. $(75, 61)$
 - d. $(66, 70)$

- 60 A store sells burgers for 8 dollars and salads for 3 dollars. If 85 items were sold for a total of 305 dollars, how many of each type were sold?
- a. (75, 10)
 - b. (15, 70)
 - c. (10, 75)
 - d. (5, 80)
- 61 A store sells adult tickets for 9 dollars and child tickets for 3 dollars. If 54 items were sold for a total of 288 dollars, how many of each type were sold?
- a. (33, 21)
 - b. (26, 28)
 - c. (16, 38)
 - d. (21, 33)
- 62 A store sells burgers for 6 dollars and salads for 2 dollars. If 79 items were sold for a total of 294 dollars, how many of each type were sold?
- a. (29, 50)
 - b. (34, 45)
 - c. (39, 40)
 - d. (45, 34)
- 63 Solve for x : $x/4 + x/6 = 20$.
- a. 60
 - b. 48
 - c. 36
 - d. 20

64 Solve for x : $x/3 + x/5 = 8$.

a. 15

b. 8

c. 1

d. 30

65 Solve for x : $x/3 + x/5 = 40$.

a. 60

b. 40

c. 75

d. 90

66 Solve for x : $x/5 + x/6 = 55$.

a. 55

b. 150

c. 120

d. 180

67 Solve for x : $x/2 + x/4 = 6$.

a. 4

b. 6

c. 8

d. 12